So far, this is entirely general. However, if the observable **L** is associated only with *A*, then it acts trivially on the *b*-index and we can write the expectation value as

$$\langle \mathbf{L} \rangle = \sum_{a,b,a'} \Psi^*(a'b) L_{a',a} \Psi(ab). \qquad (7.16)$$

Now, Alice can summarize all of her knowledge, at least for the purpose of studying *A*, in terms of a matrix ρ:

$$\rho_{aa'} = \sum_{b} \Psi^*(a'b) \Psi(ab). \qquad (7.17)$$

Surprisingly, Eq. 7.16 has exactly the same form as Eq. 7.14 for expectation value of a mixed state. Indeed, only in the very special case of a product state will ρ have the form of a projection operator. In other words, despite the fact that the composite system is described by a perfectly pure state, the subsystem *A* must be described by a mixed state.

There's a subtle point about our notation for density matrices that's worth noticing: in Eq. 7.17, the right-hand index of ρ, that is, the a' index, corresponds to the complex conjugage state-vector ψ^* ($a'b$) in the summation. This is a consequence of our convention

$$L_{aa'} = \langle a | \mathbf{L} | a' \rangle$$

for labeling the matrix elements of an operator **L**. Applying this convention to

$$\rho = |\Psi\rangle\langle\Psi|$$

results in